Assignment_1

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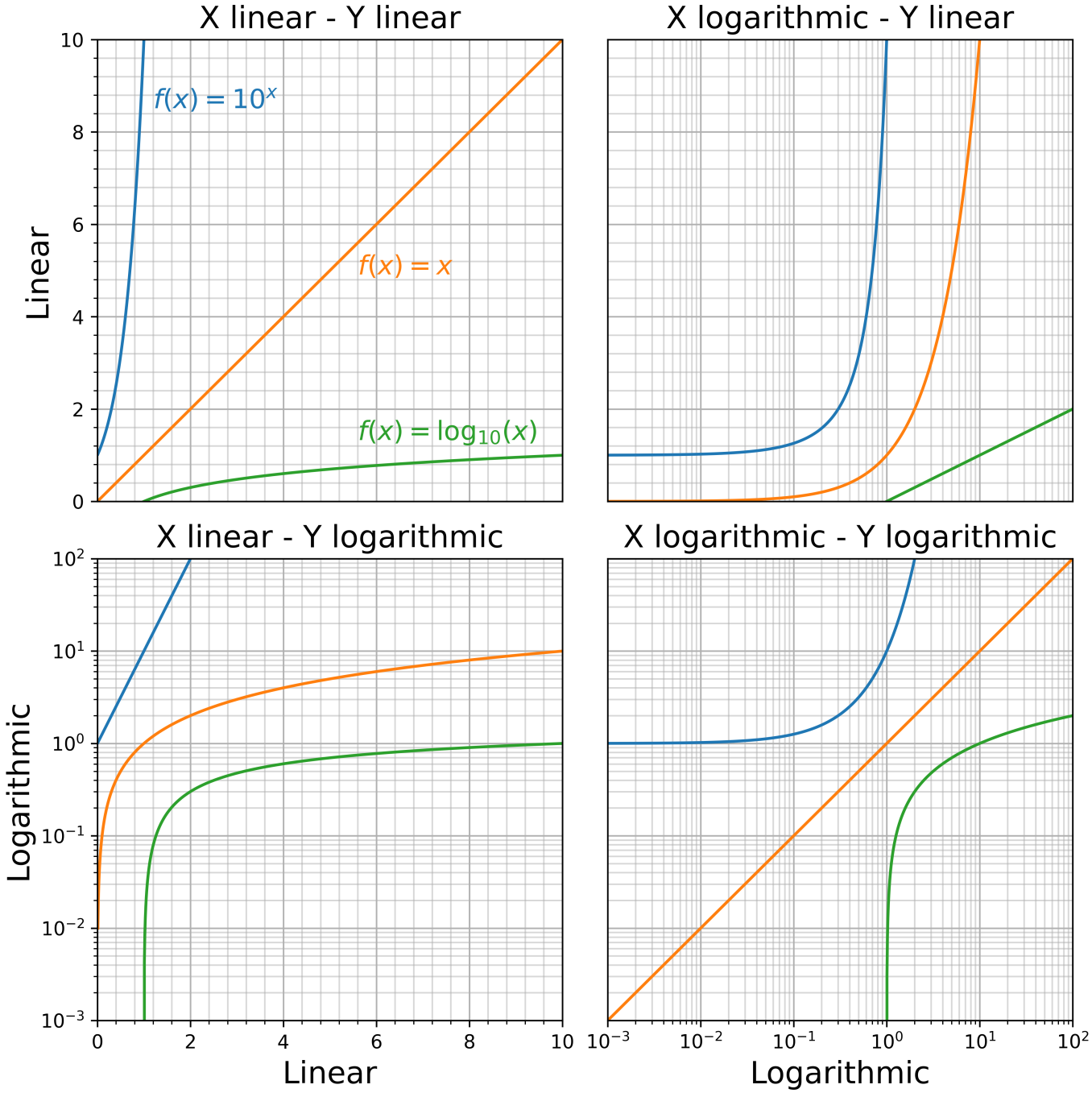
1. Reproduce the figure for Q3 in the Pre-course Quiz.

This is my resulting image:



Problem 1 code

2. Exercise 5.7 in Hsieh's book

$\boxed{5.7}$

From the given dataset containing wind direction (θ) (in degrees) and ozone concentration at a weather station in Milwaukee in 1975, perform linear regression with ozone concentration as the response variable. (a) Try different predictors in the following four models runs: (i) θ, (ii) $\cos(2\pi\theta/360)$ and $\sin(2\pi\theta/360)$, (iii) $\cos(2\pi\theta/360)$ and (iv) $\sin(2\pi\theta/360)$. (b) Add 60° to θ and repeat the four runs. Explain if there is any difference in the RMSE between the runs in (b) and the corresponding ones in (a). [Data source: R. Johnson and Wehrly (1977, table 1)]

Table of RMSE for different cases:

	a	b
i	20.396	28.220
ii	19.765	19.765
iii	28.194	23.636
iv	20.587	23.458

Caveat

When dealing with the case of $\theta' = \theta + 60$, we need to do one more operation, that is the modular operation which makes the transformed angles lie in the correct range: $[0, 360)$. More specifically, we let $\theta' = (\theta + 60) \mod 360$.

Discussion of the results

- although case a-i is not bad within case a, but using the parameter θ directly is unstable when adding a constant to it, we can see this effect in case b-i, which is the worst within case b.
- the RMSE values of case a-ii and b-ii are equivalent, this is not a coincidence, in fact, it is related to the fact that when one perform an invertible linear transformation on the predictor variables, the RMSE is invariant under such transformation. we can prove this fact in the following section.

RMSE is invariant under an invertible linear transformation on the predictor variables

Why do we care about the linear transformation on the predictor variables?

Consider the predictor variables in case ii, denote $k = 60^\circ$ and let θ being radians of the raw angular data, $\theta' = \theta + k$ being the shifted angle, then we have four predictors:

- Predictor 1 of case a: $X_1^a = \cos(\theta)$
- Predictor 2 of case a: $X_2^a = \sin(\theta)$
- Predictor 1 of case b: $X_1^b = \cos(\theta + k)$
- Predictor 2 of case b: $X_2^b = \sin(\theta + k)$

By trigonometric identities:

$$\begin{cases} X_1^b = \cos(k) \cdot X_1^a - \sin(k) \cdot X_2^a \\ X_2^b = \sin(k) \cdot X_1^a + \cos(k) \cdot X_2^a \end{cases}$$

which means we are essentially dealing with **linear transformation of the predictor variables**!

Transformation of the estimated parameters

Recall that for a model $y = X\beta$, we have:

- the estimated parameter $\hat{\beta} = (X^T X)^{-1} X^T y$
- the predicted response $\hat{y} = X\hat{\beta}$

Denote the transformed predictors $X' = TX$, in which the matrix T encapsulates the transformation rule, the model now becomes $y = X'\beta'$, and similarly:

- the estimated parater $\hat{\beta}' = (X'^T X')^{-1} X'^T y$
- the predicted response $\hat{y'} = X'\hat{\beta}'$

By matrix algebra we can establish the relation between $\hat{\beta}$ and $\hat{\beta}'$

$$\begin{aligned} \hat{\beta}' &= (X'^T X')^{-1} X'^T y \\ &= [(T^T X^T X T)^{-1}] T^T X^T y \\ &= [T^{-1} (X^T X)^{-1} (T^T)^{-1}] T^T X^T y \\ &= T^{-1} (X^T X)^{-1} X^T y \\ &= T^{-1} \hat{\beta} \end{aligned}$$

then we can show that $\hat{y'} = \hat{y}$

$$\hat{y'} = X'\hat{\beta}' = XT(T^{-1}\hat{\beta}) = X\hat{\beta} = \hat{y}$$

since $\hat{y'} = \hat{y}$, the RMSE stays the same under transformation T.

Note that the usually used standardization of predictors, say $z = (x - \bar{x})/s$ is also a linear transformation and thus it won't affect the RMSE either.

By the way, when use a single component(say $\sin$ or $\cos$) will not guarantee that RMSE being invariant under shifting of angles, since it may introduce another component, roughly speaking:

$$\sin(\theta + k) = \sin(\theta)\cos(k) + \sin(k)\cos(\theta)$$

another independent variable $\cos(\theta)$ appears, this explains the fact that in case iii and iv the RMSE is not invariant.

Problem 2 code

3. Exercise 5.8 in Hsieh's book

5.8

In this exercise, we learn how validation data help to prevent overfitting from using too many predictors. Consider the given dataset containing the winter snow water equivalent (SWE) for Grouse Mountain, BC, Canada (the winter SWE being the maximum observed SWE from January to June) and four predictors (the Niño3.4 index and the PNA, NAO and AO teleconnection indices, averaged December to February) (see Exercise 5.3). Use the first 45 data points for training and the remaining 23 points for validation. Compute the RMSE over the training data and the RMSE over the validation data for (a) the MLR model predicting SWE using all four predictors. (b) Drop the weakest predictor based on the p-values for individual predictors and repeat MLR using three predictors. (c) Again drop the weakest predictor and repeat using two predictors and (d) drop the weaker predictor and repeat using one predictor. Which MLR model would you pick based on the best validation RMSE? Would you have picked this model if you only looked at the p-values of individual predictors based on the training data?

This exercise is related to the so called "Backward Elimination", and result is summarized by the table below:

round	Nino34	PNA	NAO	AO	RMSE_train	RMSE_validation
1	0.25	0.02	0.24	0.29	424.74	544.18
2	0.25	0.01	0.57		430.9	552.96
3	0.26	0.01			432.62	541.9
4		0			439.22	578.3

Discussions

1. By assessing the model with lowest RMSE_validation, the best model would be round 3 which contains Nino34, PNA as predictors.
2. No, if I was asked to choose two predictors based on the p-values on round 1 model, I may choose PNA and NAO with lower p-values as predictors, which is not necessary the best one according to the criteria above.

Problem 3 code

4. Exercise 5.9 in Hsieh's book

$\boxed{5.9}$

In this exercise, we study the effect of collinearity among predictors and the advantage of ridge regression. Consider the given dataset containing the winter snow water equivalent (SWE) for Grouse Mountain, BC as used in Exercise $\boxed{5.8}$. Use only the first two predictors, that is, the Niño3.4 and PNA indices. Standardize the two predictors and the SWE. Next, construct a collinear predictor from the second predictor by adding a small noise term, namely $0.001\mathcal{N}(0,1)$, with $\mathcal{N}(0,1)$ being the standard Gaussian distribution, that is, the second and third predictors are very strongly correlated. Use the first 45 data points for training and the remaining 23 points for validation. Compute the RMSE over the training data and the RMSE over the validation data for the MLR model predicting SWE from the three predictors. Next perform ridge regression with the weight penalty parameter being small, $\lambda = 10^{-5}$, and being relatively large, $\lambda = 0.01$. Repeat the MLR and the two ridge regression model runs 100 times using different random noise in the third predictor. From the 100 trials, compute and compare the following statistics from the MLR and from the two ridge regression models: standard deviation for the regression parameters $\hat{a}_0, \hat{a}_1, \hat{a}_2$ and $\hat{a}_3$, the mean and standard deviation for the RMSE over training data and for the RMSE over validation data.

The result of my simulation is summarized by the table below, note that $\lambda = 0$ corresponds to MLR. Also, I have fixed the random seed in my code as `np.random.seed(666)` in order to keep the reproducibility.

statistics \ λ	**0.0**	**1e-05**	**0.01**
a0_hat_std	0.022	0.015	0
a1_hat_std	0.026	0.016	0
a2_hat_std	138.125	90.109	0.265
a3_hat_std	138.126	90.111	0.265
RMSE_train_mean	0.812	0.813	0.822
RMSE_train_std	0.013	0.01	0
RMSE_validation_mean	1.035	1.037	1.03
RMSE_validation_std	0.028	0.025	0

Discussions

1. It's easy to see that for highly colinear pairs($\hat{a_2}$, $\hat{a_3}$), the variances of these estimators are pretty high and nearly equal. This phenomenon can be explain by eigen-decomposition of a symmetric matrix(X^TX), since x_2, x_3 are nearly parallel, there exist an nearly zero eigenvalue and the corresponding eigenvector $[0,0,1,-1]^T$ such that $\mathrm{Var}\hat{\beta} \sim (X^TX)^{-1} = \sum_i (1/\lambda_i) v_i v_i^T$ would produces large value at β_2 and β_3

2. Once we've raised the regularization parameter, the variance of those colinear pairs are largely suppressed, showing the effect of regularization on the problem of co-linearity.
3. The effect of $\lambda = 10^{-5}$ is clearly not enough, since almost all statistics are nearly equal, so it's reasonable to use higher λ for highly colinear predictors. Regularization suppresses variance because it inflates the small eigenvalues of $X^T X$, preventing the inverse matrix from exploding in directions caused by collinearity.
4. Although regularization reduced the variance of the coefficient estimates, the mean RMSE for both the training and validation sets remained largely unchanged.

Problem 4 code